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GEOLOGICAL SURVEY CIRCULAR 646

Origin of the Brown Mountain Light in North Carolina

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By George Rogers Mansfield[1]

GOVERNMENT INVESTIGATIONS MADE

For many years "mysterious lights" have been seen near Brown Mountain, in the northern part of Burke County, N.C., about 12 miles northwest of Morganton. Some have thought that these lights were of supernatural origin; others have dreamed that they might indicate enormous mineral deposits; and many who have not had such visions have looked upon them as a natural wonder that lent interest to all vacation trips to the region.

In October 1913 at the urgent request of Representative E. Y. Webb, of North Carolina, a member of the U.S. Geological Survey, D. B. Sterrett, was sent to Brown Mountain to observe these lights and to determine their origin. After a few days investigation Mr. Sterrett declared that the lights were nothing but locomotive headlights seen over the mountain from the neighboring heights. This explanation was too simple and prosaic to please anyone who was looking for some supernatural or unusual cause of the lights, and when they were seen after the great flood of 1916, while no trains were running in the vicinity, even some of those who had accepted Mr. Sterrett's explanation felt compelled to abandon it.

As time went on, the interest in the lights became more general, and as one after another local investigator failed to discover their origin, the mystery seemed to grow deeper. Finally Senators Simmons and Overman prevailed upon the Geological Survey to make a second and more thorough investigation of these puzzling lights. The present writer, to whom the task of making this investigation was assigned, spent 2 weeks near Brown Mountain in March and April 1922 and took observations on seven evenings, on four of them until after midnight, from hillsides that afforded favorable views of the lights. The results of the work are reported here.

[1]Issued in 1922 as Press Notice 14328.

AID RECEIVED IN THIS INVESTIGATION

The writer gratefully acknowledges his indebtedness to Messrs. R. T. Claywell, A. M. and Charles Kistler, and H. L. Millner, of Morganton, who gave him much information and assisted him in many ways in his investigation. Joseph Loven, of Cold Spring, and H. C. Martin, of Lenoir, accompanied him on some of the evenings of observation. G. E. Moore, of Lenoir, furnished valuable information. F. H. May, of Lenoir, organized a party to accompany him to the summit of Brown Mountain and generously rendered much valuable aid. Monroe Coffey and Theodore Crump, of the U.S. Forest Service, extended to him the hospitality of their camp on Brown Mountain and joined in the investigation.

Drs. W. J. Humphreys and Herbert Lyman and Mr. C. F. Talman, of the U.S. Weather Bureau, obligingly furnished correspondence and unpublished manuscript relating to the Brown Mountain light, and W. W. Scott, of Washington, kindly lent a scrapbook containing copies of his own and other published articles relating to Brown Mountain.

The writer is also indebted to his colleagues of the Geological Survey for helpful suggestions and discussions, particularly to Arthur Keith for information about the geology of the Brown Mountain region and to R. H. Sargent, J. B. Mertie, Jr., and A. C. Spencer for aid in the interpretation of instrumental observations.

TOPOGRAPHY OF BROWN MOUNTAIN

The shape and general elevation of Brown Mountain are shown on the accompanying map. Its eastern ridge forms part of the boundary between Burke and Caldwell Counties. Its top is plateaulike and reaches a maximum elevation of about 2,600 feet. It is partly cut away by southward-flowing branches of Johns River and is separated from more intricately carved uplands on the northwest, north, and northeast by Upper and Wilson Creeks and their tributaries. Seen from a distance from almost any direction, Brown Mountain appears as a ridge having a nearly even skyline. (See map, fig. 1.)

GEOLOGIC FEATURES

The geologic features of the Brown Mountain region are the southward extension of the features seen farther north, which are described and mapped in the Cranberry folio, No. 90 of the series of folios of the Geologic Atlas of the United States. There is nothing unique or unusual in the geology of Brown Mountain. Most of the mountain is composed of the Cranberry Granite, a rock which also underlies many square miles on the north side of the Blue Ridge.

The Caldwell Power Co. has drilled a series of holes, 50 to nearly 100 feet deep, along the lower part of the east slope of Brown Mountain preliminary to the location of a tunnel. Through the kindness of H. L. Millner, an officer of the company, the writer was permitted to examine the cores taken from these holes. Most of them consisted of ordinary granite, though a few included masses of rock of other kinds. The men who surveyed the line for the tunnel reported local magnetic attraction amounting to a deflection of about 6°, but though representative pieces of all the different kinds of cores were presented to the compass needle, they produced no noticeable effect. Dip-needle tests made to determine magnetic conditions at Brown Mountain gave readings of 41½°, which is slightly greater than those made at Loven's or at Gingercake Mountain (40°) but less than those made at Blowing Rock (43°) and at the Perkins place, near Adako (45°).

RECORDS OF EARLIER OBSERVATIONS

So far as the writer is aware the first published account of the light was given in a dispatch from Linville Falls to the Charlotte Daily Observer, dated September 23, 1913, in which its discovery is credited to members of the Morganton Fishing Club, who saw it "more than two years ago" but who were "laughed at and accused of seeing things at night." This account is quoted in part below:

"The mysterious light that is seen just above the horizon almost every night from Rattlesnake Knob, near Cold Spring, on the Morganton road * * * is still baffling all investigators * * *. With punctual regular-

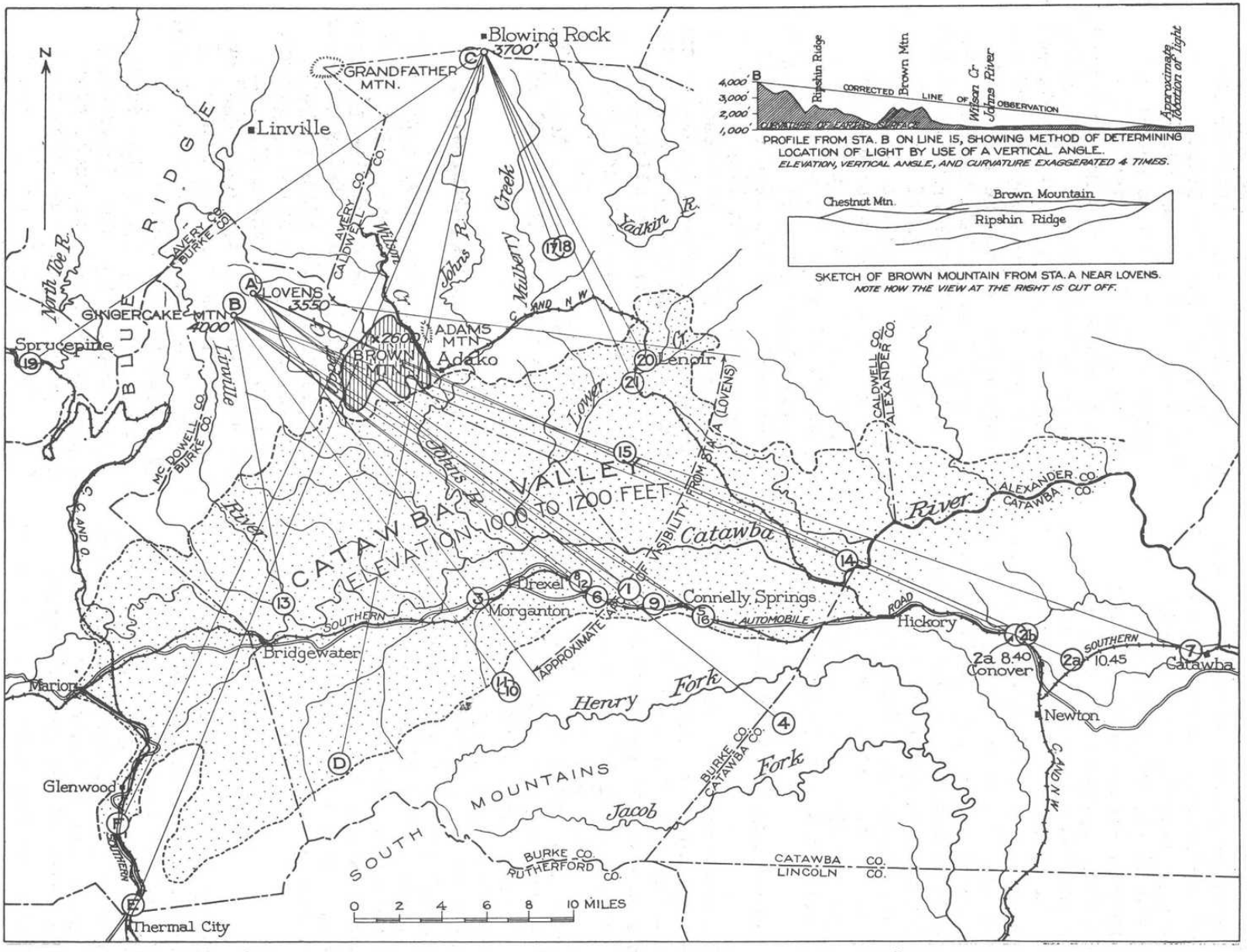


Figure 1.--Map of Brown Mountain region, N.C., illustrating origin of "Brown Mountain lights."

ity the light rises in a southeasterly direction from the point of observation just over the lower slope of Brown Mountain, first about 7:30 p.m. and again at 10 o'clock. It looks much like a toy fire balloon, a distinct ball, with no atmosphere about it. * * * It is much smaller than the full moon, much larger than any star and very red. It rises in the far distance from beyond Brown Mountain, which is about 6 miles from Rattlesnake Knob, and after going up a short distance, wavers and goes out in less than 1 minute * * *. It does not always appear in exactly the same place, but varies what must amount in the distance to several miles. The light is visible at all seasons, so Mr. Anderson Loven, an old and reliable resident testifies * * *. There seems to be no doubt that the light rises from some point in the wide, level country between Brown Mountain and the South Mountains, a distance of about 12 miles, though it is possible that it rises at a still greater distance."

In this article in the Charlotte Daily Observer, the discovery of the lights is assigned to a date "more than 2 years ago," but conversation with B. S. Gaither, of Morganton, who participated in the fishing party mentioned, and who was the one who first saw the lights, elicited the fact that they were observed in 1908 or 1909. Rev. C. E. Gregory, who in 1910 built a cottage on the little knoll near Loven's Hotel at Cold Spring, presumably the Rattlesnake Knob referred to, was, according to local oral accounts, the first to give much attention to the lights and to bring them to public notice.

H. L. Millner, an engineer living at Morganton, states that he did railroad surveying all through the mountains north of Brown Mountain in 1897, 1899 to 1903, and in 1905. He afterward spent many summer vacations in those mountains but says that he never saw the lights and never heard of them until 1910. Similar testimony is given by Rev. Albert Sherrill, who served two churches in the country about Brown Mountain. In a letter to Dr. W. J. Humphreys dated January 25, 1922 he says:

"For 4 years I traveled the roads to these churches and visited in the homes of the people all about this mountain. I held revival services day and night, which gave me a chance of observation at night. This was from 1909 to 1912, inclusive. At no time in all these years did I <u>see</u> a light or <u>hear</u> of one * * *. Two years after I left there was the first I ever heard of it."

On the other hand, Col. Wade H. Harris, editor of the Charlotte Daily Observer, from which the first description of the light, quoted above, was taken, states in a letter dated October 2, 1921, addressed to Senator Simmons, that "there is a record that it (the light) has puzzled the people since and before the days of the Civil War." R. T. Claywell, of Morganton, says that people used to come to Burke County 60 years ago to see the lights. Joseph Loven, of Cold Spring, says that he noticed the lights as early as 1897, when he moved to his present home by Loven's Hotel, but that he had heard nothing about them and paid no attention to them until Mr. Gregory came, in 1910.

In October 1913, Mr. Sterrett of the U.S. Geological Survey made his investigation, as a result of which he decided that the lights were locomotive headlights. He did not visit Rattlesnake Knob but went unaccompanied to the Brown Mountain region, where he made his observations.

A newspaper article on the lights, by W. W. Scott, published November 10, 1915, aroused much local interest and started newspaper discussions, as a result of which several expeditions, made up of local observers, visited

Brown Mountain early in the winter of 1915, and in the spring of 1916 and attempted to determine the character and source of the lights. The members of these expeditions made some interesting observations but did not satisfactorily achieve their object.

Mr. H. C. Martin, of Lenoir, states that on April 11, 1916, he and Dr. L. H. Coffey organized an expedition to study the Brown Mountain light. Mr. Martin's party went to Adams Mountain. Dr. Coffey's party went to Brown Mountain. Each party subdivided into several groups and signals were arranged that whichever group first saw the light should fire a pistol. Dr. Coffey's party saw the light over the summit of Adams Mountain at 8:10 and again at 9:45, over a point somewhat farther south. About 5:10 a.m. they saw the light again over the south end of Adams Mountain. None of these appearances was seen by Mr. Martin's party, but about 11:52 his party saw two lights (floating globes), "apparently about the size of ordinary street lamps of Lenoir seen from the distance of about 1 mile," flash out among the trees on the east side of Brown Mountain about one-eighth of the distance down from the summit. These lights moved horizontally southeastward, floating in and out of the ravines, along the mountainside past a dead pine tree in Mr. Martin's line of sight for a distance estimated at 2 miles. Then they returned northwestward about half that distance, again passing the line of the dead tree. At 12:13 the lights disappeared as suddenly as they came. These lights were not seen by Dr. Coffey's party.

In the summer of 1916, a great flood swept down the valley of Catawba River, washing out bridges and railroad tracks and suspending all railroad traffic in and about Morganton, so that for several weeks no trains came within 40 or 50 miles of Rattlesnake Knob, yet during that period the lights were reported to be seen as usual. This fact showed that the Brown Mountain lights could not be ascribed solely to locomotive headlights.

Late in 1919 the question of the origin of the Brown Mountain light was brought to the attention of the Smithsonian Institution and referred to the U.S. Weather Bureau. Descriptions given in letters from trustworthy observers led Dr. W. J. Humphreys, of that Bureau, to decide that the light was an electrical discharge analogous to the "Andes light" of South America. This Andes light and its possible relation to the Brown Mountain light became the subject of a paper presented by Dr. Herbert Lyman before a meeting of the American Meteorological Society held at the Weather Bureau in Washington in April 1921. Soon thereafter the suggestions of the physicists of the Weather Bureau were embodied in a bulletin on the Brown Mountain light issued by the National Geographic Society, in which this light was represented as a manifestation of the Andes light. Neither the Weather Bureau nor the National Geographic Society, however, had sent an investigator to Brown Mountain to observe the lights.

REVIEW OF EARLIER OBSERVATIONS

Those who have seen the lights from the south or east may with justice contend that no locomotive headlights can be seen to the north and northwest. A good topographic map, however, shows many roads on which an automobile headlight might intervene between an observer and Brown Mountain in such a way as to give much the same effect that one would get in viewing it over the mountain from Loven's or Blowing Rock.

There are two buildings on the summit of Brown Mountain. One of these

is owned by the Brown Mountain Club and the other is a Forest Service station. Lights in these and the fires of campers are the only lights known to originate on Brown Mountain and are the only lights that have been seen on the mountain by observers on the mountain. There are also some buildings on the southern spurs of Brown and Adams Mountains. Seen from favorable viewpoints these might furnish lights that could be interpreted as manifestations of the Brown Mountain light.

For some years there have been lumber camps in Upper Creek west of Brown Mountain. Some of the buildings at these camps are on the sides of the valley or on Ripshin Ridge. Viewed from favorable positions lights in these buildings might be ascribed to the Brown Mountain light.

Rare electrical discharges similar to those reported from western Virginia may take place on Brown Mountain or on other ridges on the Blue Ridge front, but they could have nothing in common with the ordinary Brown Mountain light either in appearance or in regularity of occurrence.

The lights seen by Mr. Martin from Adams Mountain can probably not be satisfactorily explained after so long a lapse of time. There is no reason to attribute to them supernatural or unusual origin, and they cannot be explained as due to mirage, which is Mr. Martin's idea. The suggestion that they might have been caused by moonshiners carrying lanterns has been rejected because of the roughness of the east side of Brown Mountain and because of the distance that the lights seemed to travel in 20 minutes. They might be due to fireflies flying relatively near Mr. Martin yet appearing unduly large because his eyes were focused on the distant hillside, the appearance of going in and out of ravines being due to intermittence in the lights, but H. S. Barber, an entomologist of the Division of Insects of the National Museum, to whom the matter was referred, states that this explanation, though possible, is improbable, chiefly because of the lateness of the hour of observation.

PLACES OF OBSERVATION

In the article published in the Charlotte Daily Observer in September 1913, it was stated that the lights were first observed from a place near Loven's Hotel, at Cold Spring, and as late as November 1915, Mr. Scott, in his newspaper article cited above, states that "Mr. Loven's is the only place from which has been seen the mysterious light that rises over Brown Mountain." By the spring of 1916, however, the lights were being observed from several places near Lenoir. More recently they were seen from Blowing Rock, which is visited by many tourists, who find accommodations at several flourishing hotels. The lights furnish one of the many attractions afforded by this remarkably well situated and delightful little town. Among the other localities from which the Brown Mountain lights are said to have been seen are the slopes of Gingercake Mountain, about a mile and a half southwest of Cold Spring, and the toll gate on the Yonahlossee road, on the south slope of Grandfather Mountain. It is also reported that they have been seen from Morganton and from other points in the valley southeast of Brown Mountain.

It is significant that, though many persons have from time to time camped on Brown Mountain and have spent nights watching for the lights, yet, so far as the writer has been able to ascertain, no one has actually observed the light on Brown Mountain when he himself was on that mountain,

located as to show as well as this one the atmospheric phenomena described, and the opportunities here for the observation of such phenomena are perhaps no less exceptional. Loven's Hotel and Blowing Rock, which are resorts that attract fishermen or tourists, are among the most favorable places of observation. The valley is fairly well settled and has a network of roads, three railroads, and several large towns, so that the possible sources of light are very numerous.

As the basin and its atmospheric conditions antedate the earliest settlement of the region, it is possible that even among the first settlers some favorably situated light may have attracted attention by seeming to flare and then diminish or go out. As the country became more thickly settled the number of chances for such observations would increase. Before the advent of electric lights, however, it is doubtful whether such observations could have been sufficiently numerous to cause much comment, though some persons may have noted and remarked upon them.

According to local estimates electric lights have been in use in the larger towns of the region for about 30 years. Lights from those towns viewed from the locality near Loven's, which for a long time was the only locality from which the lights were observed, are, with the exception of those from Morganton and Blowing Rock, all seen over Brown Mountain; hence the "prescribed circle" mentioned by Colonel Harris.

The use of powerful electric headlights on railway locomotives, which began about 1909, furnished new sources of strong lights in the valley and introduced an element of regularity in their appearance, which may account for the "punctual regularity" noted in the first description. After Rev. C. E. Gregory took up his summer residence near Loven's Hotel, in 1910, the Brown Mountain light began to acquire notoriety.

Meanwhile, automobiles were coming into use throughout the country, and many of them were equipped with powerful headlights. Within the last few years their number has been greatly increased, and this fact is in keeping with the general deduction already made--that on a favorable evening the lights are seen more frequently now than formerly.

During the flood of 1916, when train service was temporarily discontinued, the basin east of Marion, where the atmospheric conditions are disturbed, was still the scene of the intermittent flare of favorably situated lights. Automobiles were then in use in the larger towns and on some of the intervening roads, and their headlights were doubtless visible from Loven's over Brown Mountain. One need only remember the network of roads in the valley region (see topographic maps of the Morganton and Hickory quadrangles) to realize the almost infinite number of possibilities for automobile headlights to be pointed toward Brown Mountain and stations of observation beyond. It should be emphasized, too, that automobile headlights and locomotive headlights, when seen at distances and under atmospheric conditions such as those which prevail in this region, possess no characteristic that clearly distinguishes them from other lights. On the contrary, as stated by the lady at Blowing Rock, they look "as much alike as so many peas in a pod," though this statement should not be understood to mean that some may not be brighter than others.

Col. Wade H. Harris, in his letter to Senator Simmons already cited, says: "A locomotive headlight is easily and unmistakably distinguished as such, not only by the rays it shoots forth, but from its movements * * *.

element of motion has already been considered.

The arc of view intercepted by Brown Mountain is a little greater than the arc between the lines D and F on the accompanying map (fig. 1). Line D is drawn to a road that seems to be a thoroughfare near Brindletown. Line E is drawn tangent to the Southern Railway near Thermal City. Just east of this line is a thoroughfare, and a short distance farther east is the track of the Carolina, Clinchfield & Ohio Railway. Line F is drawn to the same road and railroads a few miles farther northwest.

Automobile headlights on any of the numerous roads that point toward Blowing Rock within the area designated and south of Catawba River would be visible from Blowing Rock over Brown Mountain.

The writer was told at Blowing Rock that a good time to see the light was from 9:30 p.m. to 10 or a little later. The agent of the Southern Railway at Thermal City states that a northbound train on that railway passes that station at 9:30 p.m., maintaining approximately the same schedule the year round. The agent at Glenwood on the same railway reports that a train is due there at 9:53 the year round. Data for the other railroad are not available, but doubtless there are some northbound evening freight trains on it. The agreement of the train schedule with the above statement about the time to see the lights is certainly more than a coincidence.

It may be questioned whether a locomotive headlight could be seen for a distance so great as that between Blowing Rock and Thermal City, which is about 45 miles. The Minots Ledge light, already mentioned, is rated at 75,000 candlepower by the Bureau of Lighthouses and is visible for distances greater than 25 miles. The writer was told by Mr. Chadwick, of the Engineering Department of the Southern Railway, that the headlights in common use on that system are incandescent nitrogen lamps rated at 250 watts and 32 volts. Fitted with 16-inch silvered-copper parabolic reflectors, these lights yield about 600,000 candlepower. There is therefore no reason to doubt that the headlights would be visible at a distance of 45 miles.

The high power of these lights accounts for the brilliancy ascribed to the Brown Mountain light by observers who have seen it when the air was exceptionally clear, and it also accounts for the fact that some of the lights seen are brighter than others.

In summary it may be said that the Brown Mountain lights are clearly not of unusual nature or origin. About 47 percent of the lights that the writer was able to study instrumentally were due to automobile headlights, 33 percent to locomotive headlights, 10 percent to stationary lights, and 10 percent to brush fires.

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